

Data Tidying

Getting the data to a workable state

`dplyr` is a grammar
for tidying data.

[Here](#) is a cheatsheet of
commands.

Types of Headaches

Migraine



Hypertension



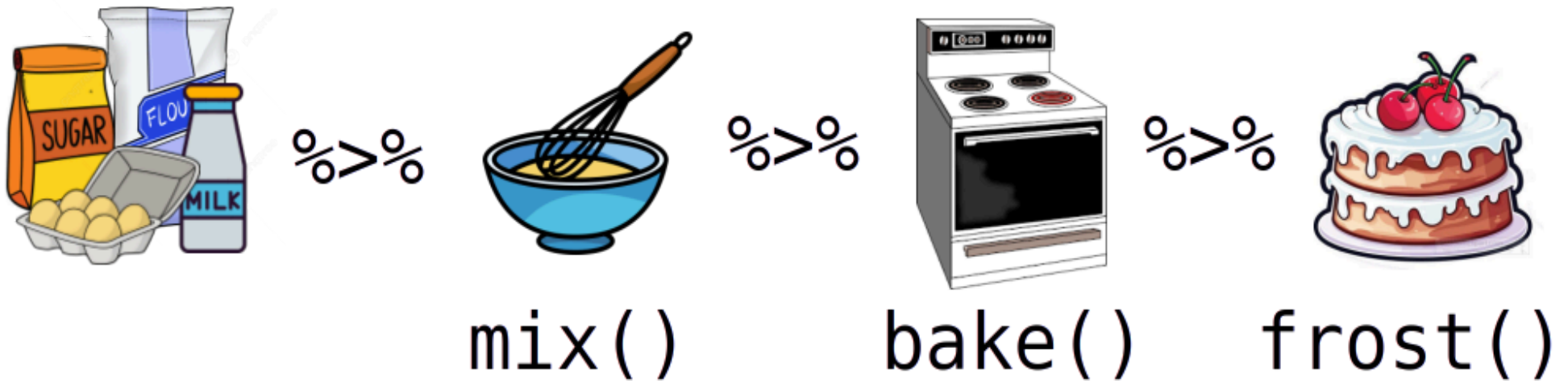
Stress



Handling data issues



The pipe (`%>%`) lets us chain commands together.



Add with **Ctrl/Cmd + Shift + M**

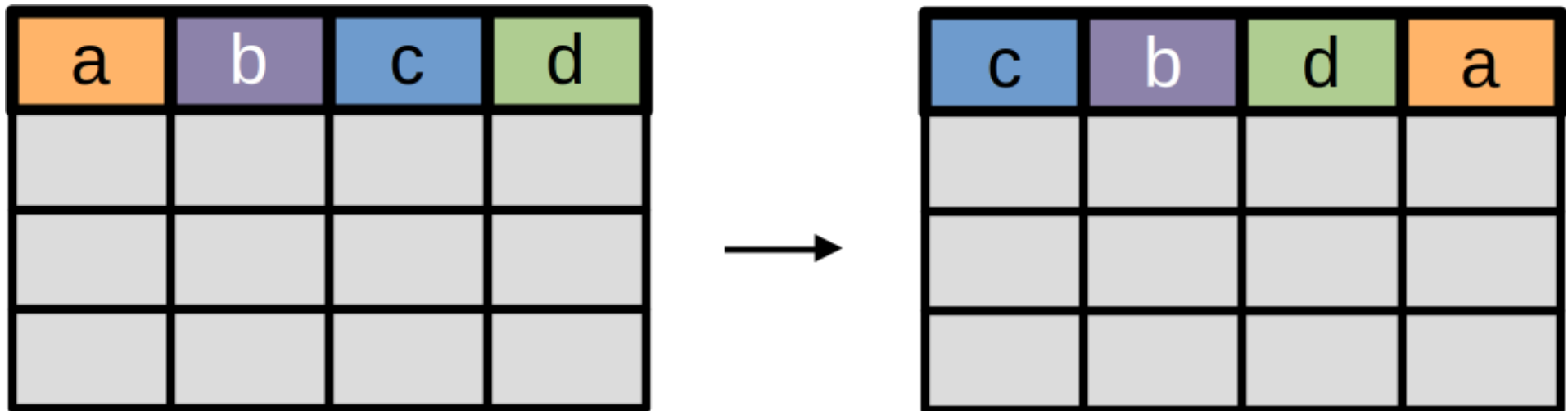
Column commands act on the columns and leave rows unchanged.

- `select()`, `relocate()`, `rename()`, `mutate()`

Row commands act on the rows and leave columns unchanged.

- `arrange()`, `filter()`, `drop_na()`

`relocate()` moves columns.



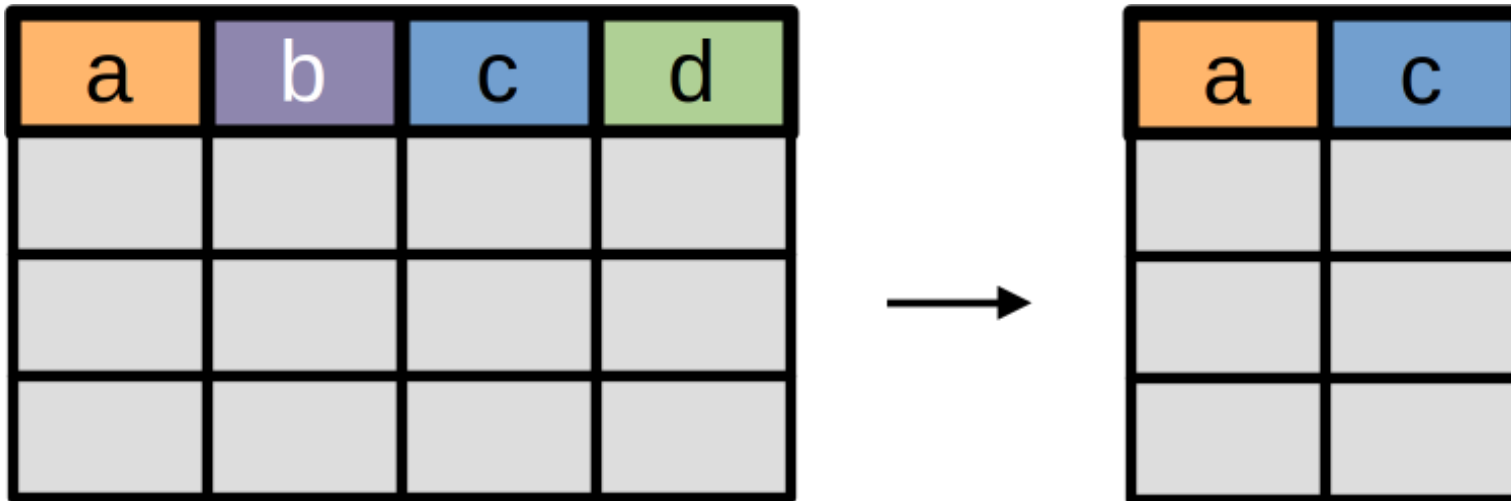
`rename()` renames columns.

a	b	c	d

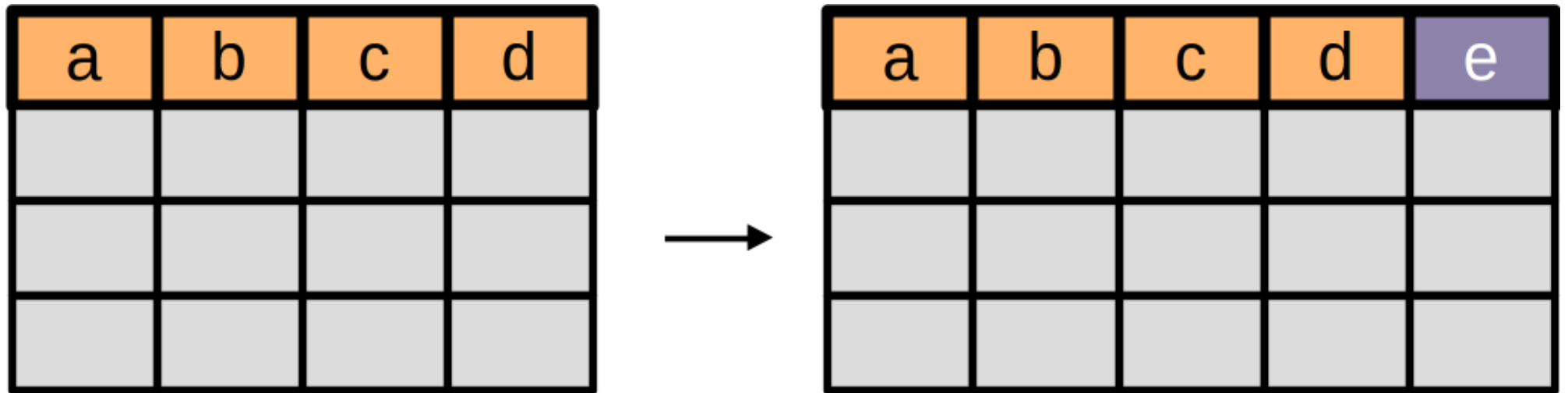


i	ii	iii	iv

`select()` reduces to specific columns.

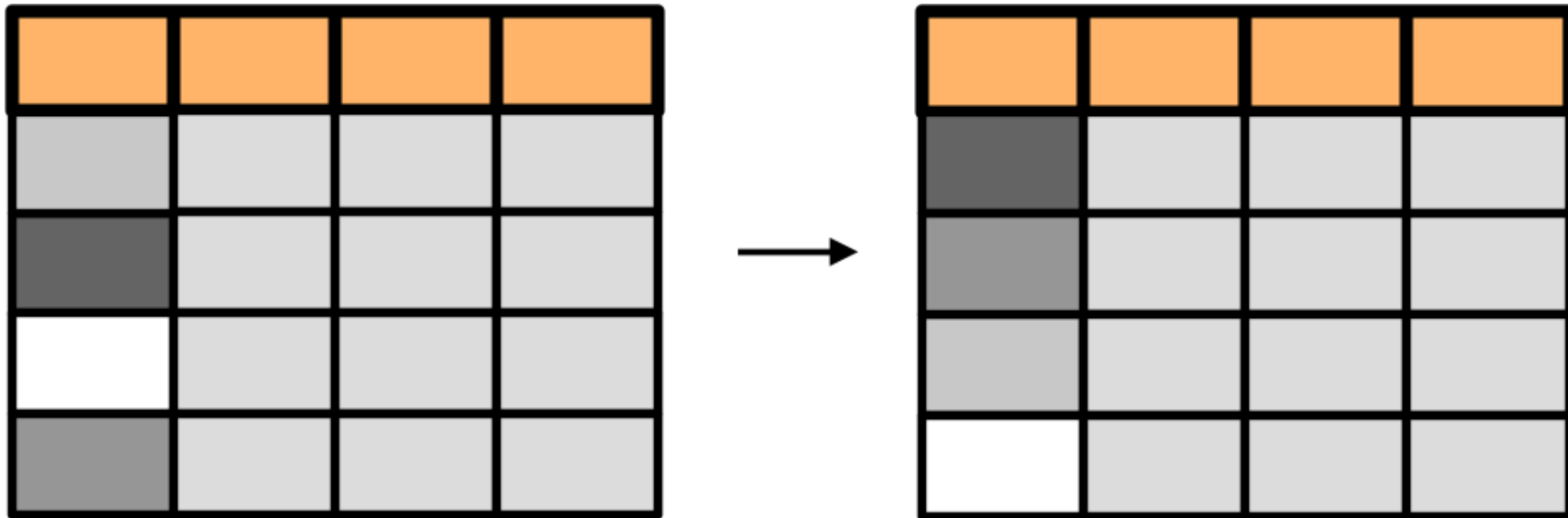


`mutate()` creates a new column.

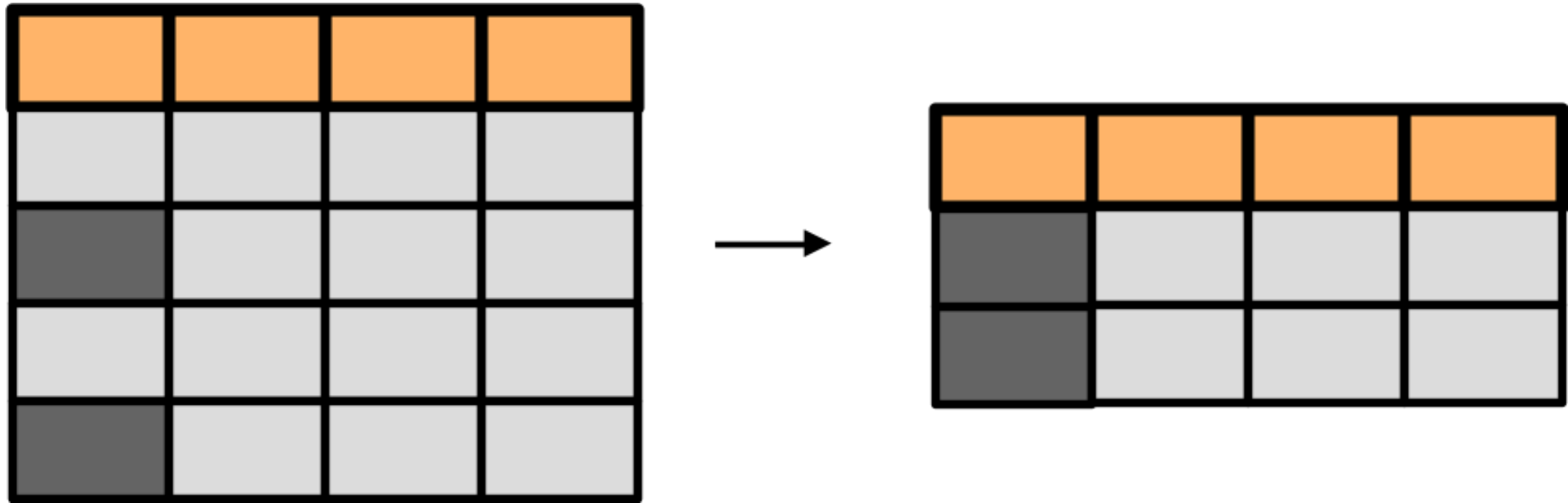


Often used with a math equation or with `case_when()`.

`arrange()` sorts rows.

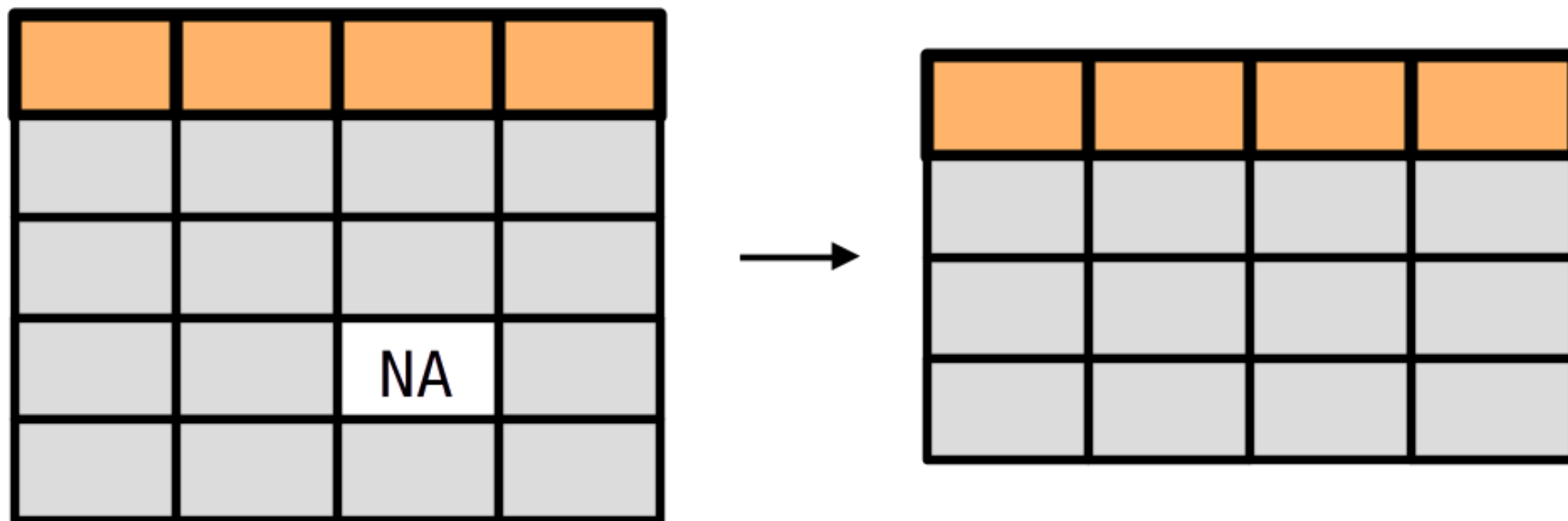


`filter()` reduces to specific rows.



Similarly, `slice_min()` and `slice_max()` return rows with the smallest or largest value of a column.

`drop_na()` removes rows with missing values.



Friends A and B are fighting over who is the most die-hard Star Wars fan. Settle the debate with the built-in `starwars` data.

- First, investigate the data. Make sure you're familiar with each column and its type.
- Write R code to resolve each of the arguments.
- Each solution will require a chain of different dplyr commands, connected with a pipe.



`summarize()` reduces a column to a single value.

- e.g. sum, min, mean, max

Combine with `group_by()` for more complex operations.

`summarize():`

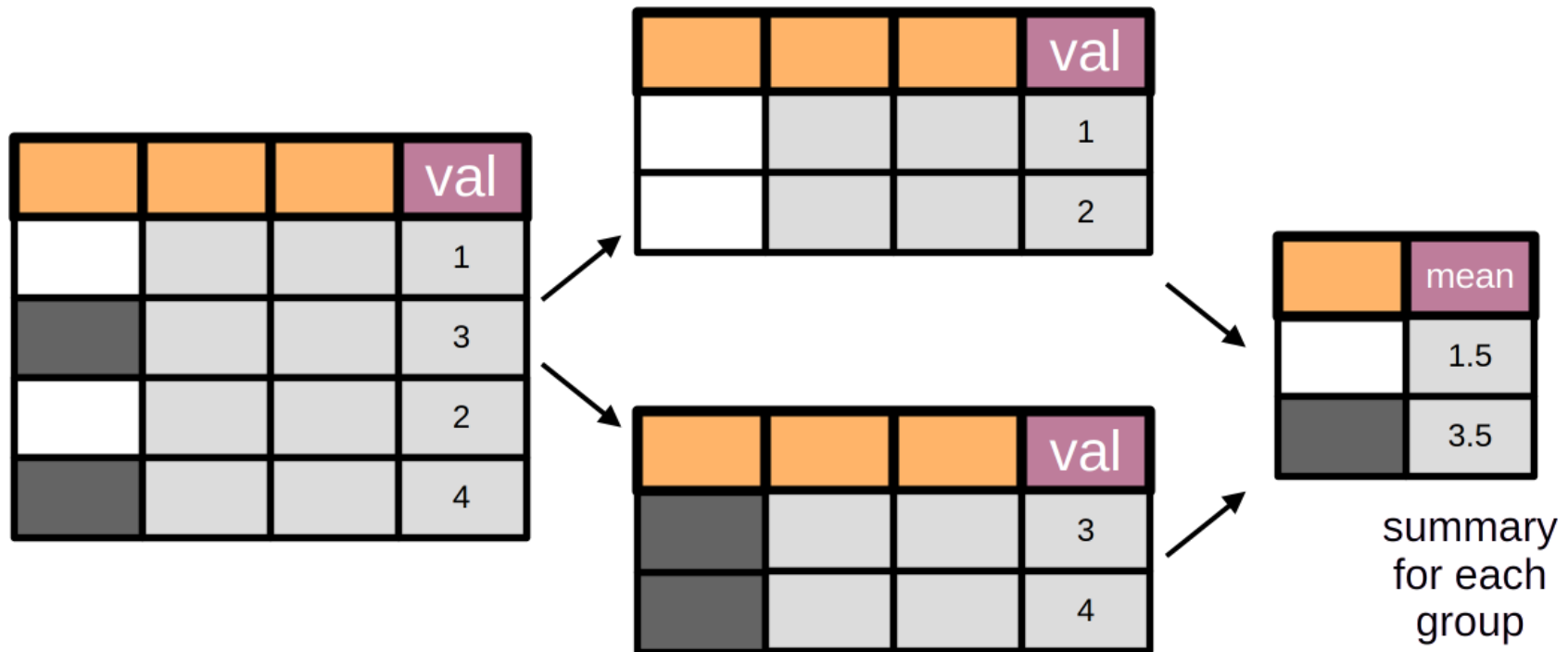
			val
			1
			3
			2
			4



mean
2.5

Single
summary
value

```
summarize() %>% group_by():
```



We can use `mutate()` with `group_by()` for sophisticated column creation.

How does each penguin's weight compare to all the others?

versus

How does each penguin's weight compare to the others of its species?



The volume of a Gentoo or Chinstrap penguin's beak is approximately $(\pi \times r^2 \times \text{length})/2.5$ and the volume of an Adelie penguin's beak is approximately $(\pi \times r^2 \times \text{length})/2$.

- Calculate each penguin's approximate beak volume (pay attention to radius vs depth!)
- Find the penguin with the largest beak volume for each combination of species and island.
- Were most of the winners measured in 2007 or 2009?

`tidyverse` also has commands to join datasets together.



Let's join `grocery_list.csv` and `grocery_prices.csv`.

Mutating joins append columns together.

- `left`, `right`, `full`, `inner`

Filtering joins keep or remove rows based on the second df.

- `semi`, `anti`

Both of these lines of code work the same:

```
join(df1, df2)           df1 %>%  
                           join(df2)
```

Also consider the *order* of joining and which df is *prioritized*.

Best practice is to specify the `join_by` argument explicitly.

`join_by` solves issues with:

- Code readability
- Columns with same name but different meanings
- Columns with same meaning but different names

Join the dataframes `drinks` and `coupons` to see which coupons can be used at a user's favorite cafe. Beware of column names!

- Use a mutating join that keeps the user's favorite drinks.
- Be very careful about what columns are being used to join.
- Mutate a new column to calculate each drink's discounted price.



Looking ahead

In [chapter 6](#), we'll learn how to use R for simulation and resampling. If you want a head start, run the following lines of code to investigate the mean flipper length of Adelie penguins:

```
1 adelic <- penguins %>% filter(species == "Adelie")
2 mean(adelic$flipper_length_mm)
3 n <- length(adelic)
4
5 new <- sample_n(adelic, size = n, replace = TRUE)
6 mean(new$flipper_length_mm)
```

What does the object `new` look like? Why does the mean of the “`new`” flipper length change every time you run the code?